

Stress intensity factors at tips of branched cracks under various loadings

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Received 10 June 1990; accepted in revised form 24 June 1991

Abstract. Several papers have been published on branched cracks by using various analytical methods, but most of them are concerned with special crack geometries or special loading conditions, and often give unreliable values for cracks with short branches or with small branching angles. The purpose of this paper is to give reliable formulae and new results of the stress intensity factors of various branched cracks in a wide plate. The analysis is based on the body force method combined with a perturbation procedure, and the stress intensity factors at the tips of all the branches and the main crack are given by power series formulae. Numerical results for typical branched cracks are discussed.

1. Introduction

Several papers have been published on branched cracks by using various analytical methods [1–9]. But they are concerned with special crack geometries or special loading conditions, and often give unreliable values for cracks with short branches or with small opening angles.

The purpose of this paper is to give reliable formulae and new results of the stress intensity factors at the tips of branched cracks in a wide plate under various loads at infinity. The analysis is based on continuous distributions of body force doublets along the branches. These body forces were derived from the solution of an isolated force in a plate with a line crack, and do not disturb the traction-free conditions along the main crack at all. The density functions of the applied body forces can thus be determined only from the boundary conditions along the branches.

In numerical calculation, the branches are divided into elements, the unknown body force densities are expressed by piecewise linear functions, and they are numerically determined from the boundary conditions of the branches. At this stage, the traction-free relations of the branches are replaced by the conditions for the resultant forces of all the branch elements, and this new procedure considerably improves the accuracy of the numerical results. Furthermore, a perturbation method is employed, and the stress intensity factors at the tips of all the branches and the main crack are given by power series formulae. Numerical results for various symmetrical and unsymmetrical branched cracks are discussed.

2. Theoretical analysis

2.1. Description of the problem and basic equations

Consider a branched crack in a wide plate as shown by Fig. 1. It consists of the main crack BO with length $2a$ and N branches OA_j with lengths $2b_j$ and inclined to the main crack by β_j ($j = 1, 2, \dots, N$). The coordinate system Oxy is defined with its origin at the branching point O.

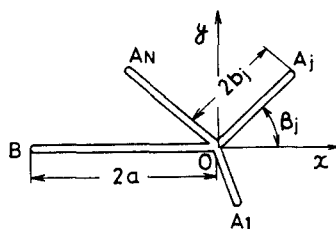


Fig. 1. Geometry of arbitrarily branched crack.

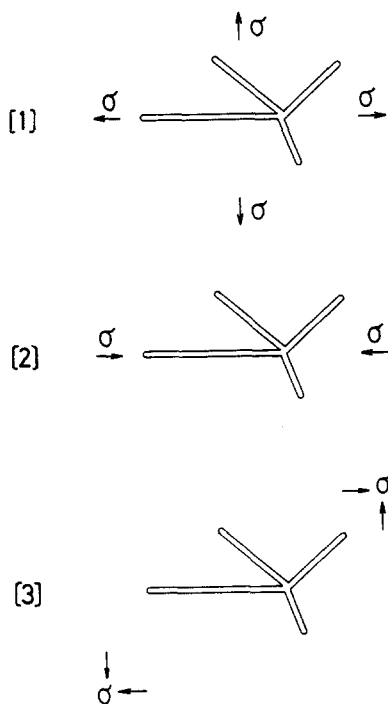


Fig. 2. Three fundamental loads.

We treat the following three fundamental loads at infinity as shown by Fig. 2, where σ is a reference stress:

1. uniform all-around tension: $\sigma_x^\infty = \sigma_y^\infty = \sigma$,
2. uniform compression in the x -direction: $\sigma_x^\infty = -\sigma$,
3. uniform shear: $\tau_{xy}^\infty = \sigma$.

The analyses are performed by assuming the traction-free conditions along all the branches of the crack. In some cases depending on the load and crack geometry, K_I at some crack tips may be negative. In these cases, the edges of the corresponding branches contact, and the assumed traction-free conditions are meaningless, at least for these specific problems. However, they will still be useful, since, for some combined loads, valid solutions with non-negative K_I values can be obtained by superposing the results for the fundamental loads.

In plane problems of elasticity, stress components σ_x , σ_y , τ_{xy} and resultant force components R_x , R_y are given in terms of two complex potentials $\Phi(z)$ and $\bar{\Omega}(z)$ as follows:

$$\left. \begin{aligned} \sigma_y + \sigma_x &= 2[\Phi(z) + \bar{\Phi}(\bar{z})] \\ \sigma_y - \sigma_x + 2i\tau_{xy} &= 2[(\bar{z} - z)\Phi'(z) - \Phi(z) + \bar{\Omega}(z)] \end{aligned} \right\} \quad (1)$$

$$R_y + iR_x = - \int \bar{\Phi}(\bar{z}) d\bar{z} - \int \bar{\Omega}(z) dz + (z - \bar{z})\Phi(z). \quad (2)$$

2.2 Resultant forces due to isolated force doublets

The analysis is performed by generalizing the previous analysis of kinked cracks [10] based on the body force method [11] combined with a perturbation procedure. We distribute proper body forces along the branches, and determine their densities so that these body forces, together with the applied loads at infinity, satisfy the traction-free conditions along the branches. These conditions are expressed in terms of resultant forces in order to improve the accuracy of the numerical results [7], and we begin with the expressions of the resultant force of a wide plate containing the main crack BO only and subjected to the same fundamental loads as shown in Fig. 2. They are derived from the known complex potentials for these problems by using (2), and are expanded in negative power series of a , regarding the later use of a perturbation procedure. Thus we have the following expressions of the resultant forces of a plate with the line crack BO subjected to the fundamental loads at infinity:

$$\left. \begin{aligned} \dot{R}_y^{[k]} + i\dot{R}_x^{[k]} &= \sigma\sqrt{\pi a} \sum_{m=0}^{\infty} \dot{R}_m^{[k]}(z) a^{-m_1} \\ m_1 &= \begin{cases} m & (k=1, 3) \\ m + \frac{1}{2} & (k=2) \end{cases} \end{aligned} \right\}, \quad (3.1)$$

where superscripts $[k]$ ($k=1, 2, 3$) denote the three fundamental loads as shown by Fig. 2, and the expansion coefficients $\dot{R}_m^{[k]}(z)$ are given as follows:

$$\left. \begin{aligned} \dot{R}_m^{[1]}(z) &= \frac{1}{\sqrt{2\pi}} \frac{(-1)^m}{2^{3m}} \frac{1}{(2m-1)} \binom{2m}{m} \left[z^{m+(1/2)} + \bar{z}^{m+(1/2)} + (2m+1) \frac{\bar{z}-z}{2} z^{m-(1/2)} \right] \\ \dot{R}_0^{[2]}(z) &= -\frac{1}{\sqrt{\pi}} \frac{z-\bar{z}}{2}, \quad \dot{R}_m^{[2]}(z) = 0 \quad (m \geq 1) \\ \dot{R}_m^{[3]}(z) &= \frac{i}{\sqrt{2\pi}} \frac{(-1)^m}{2^{3m}} \frac{1}{(2m-1)} \binom{2m}{m} \left[z^{m+(1/2)} + \bar{z}^{m+(1/2)} + (2m+1) \frac{\bar{z}-z}{2} z^{m-(1/2)} \right] \end{aligned} \right\}. \quad (3.2)$$

The above expressions (3.1) must be eliminated by the resultant forces due to the body forces to be applied along the branches. It is desirable that these body forces do not disturb the traction-free relations of the main crack. By doing this, the unknown densities of the body forces can be determined only from the boundary conditions along the branches, and highly accurate results would be expected.

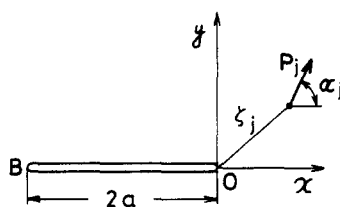


Fig. 3. Isolated force in plate with line crack.

In order to establish the body forces of this kind, we consider a wide plate with a line crack BO and subjected to an isolated force $P_j e^{i\alpha_j}$ at a point ζ_j as shown by Fig. 3. The complex potentials for this problem are written as follows [12]:

$$\Phi_j(z) = \Phi_{j0}(z) + \Phi_{j1}(z), \quad \Omega_j(z) = \Omega_{j0}(z) + \Omega_{j1}(z), \quad (4.1)$$

where $\Phi_{j0}(z)$, $\Phi_{j1}(z)$, $\Omega_{j0}(z)$, $\Omega_{j1}(z)$ are given as

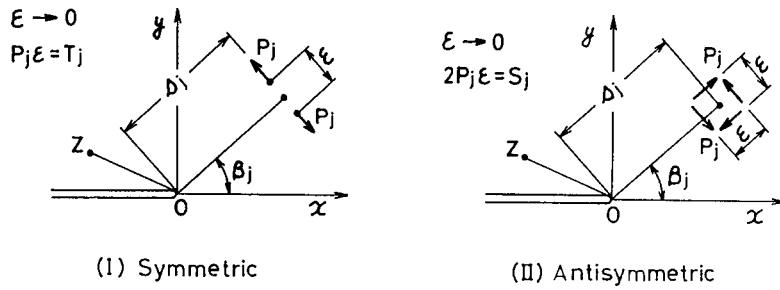
$$\left. \begin{aligned} \Phi_{j0}(z) &= -\frac{Q_j}{z - \zeta_j}, \quad \Omega_{j0}(z) = \frac{3Q_j}{z - \zeta_j} - \frac{\bar{Q}_j(\zeta_j - \bar{\zeta}_j)}{(z - \zeta_j)^2} \\ \Phi_{j1}(z) &= \Omega_{j1}(z) = \frac{Q_j}{2(z - \zeta_j)} \left[1 - \sqrt{\frac{\zeta_j(\zeta_j + 2a)}{z(z + 2a)}} \right] - \frac{3Q_j}{2(z - \bar{\zeta}_j)} \left[1 - \sqrt{\frac{\bar{\zeta}_j(\bar{\zeta}_j + 2a)}{z(z + 2a)}} \right] \\ &\quad + \frac{\bar{Q}_j(\zeta_j - \bar{\zeta}_j)}{2} \left[-\frac{1}{z - \bar{\zeta}_j} \frac{\bar{\zeta}_j + a}{\sqrt{z(z + 2a)} \sqrt{\bar{\zeta}_j(\bar{\zeta}_j + 2a)}} \right. \\ &\quad \left. + \frac{1}{(z - \bar{\zeta}_j)^2} \left(1 - \sqrt{\frac{\bar{\zeta}_j(\bar{\zeta}_j + 2a)}{z(z + 2a)}} \right) \right] + \frac{Q_j}{\sqrt{z(z + 2a)}} \end{aligned} \right\}. \quad (4.2)$$

$$Q_j = \frac{P_j e^{i\alpha_j}}{8\pi}$$

In the above, Poisson's ratio ν is assumed to be zero for simplicity, since the stress distribution for this kind of problems under stress boundary conditions is independent of the elastic constants.

Now we consider the force doublets of symmetric and antisymmetric types acting at the point $\zeta_j = s_j e^{i\beta_j}$ on the j th crack as shown by Fig. 4, (I) and (II), respectively. The complex potentials for these force doublets are derived by superposing the above expressions (4.1) with (4.2), and by taking the limiting cases when $\varepsilon \rightarrow 0$ and $P_j \rightarrow \infty$, holding $P_j \varepsilon = T_j$ or $P_j \varepsilon = S_j/2$, where T_j and S_j are the intensities of the force doublets of the symmetric and antisymmetric types. Using these functions in (2), the resultant force due to the isolated force doublets, T_j and S_j , is obtained in the following form:

$$[R_y + iR_x]_j = T_j \sum_{m=0}^{\infty} G_{j,m}(z, s_j) + S_j \sum_{m=0}^{\infty} H_{j,m}(z, s_j), \quad (5.1)$$

Fig. 4. Isolated force doublets acting on j th branch.

where the coefficients $G_{j,m}(z, s_j)$ and $H_{j,m}(z, s_j)$ are given as follows:

For $m = 0$:

$$\begin{aligned}
 G_{j,0}(z, s_j) = & \frac{1}{16\pi} \left[-2e^{-i\beta_j} \left(\frac{1}{s_j - w_1} - \frac{1}{s_j - w_2} \right) + 2i \sin \beta_j \left(\frac{e^{-i\beta_j}}{s_j} \right)^{1/2} \left(\frac{z^{1/2}}{s_j - w_1} + \frac{\bar{z}^{1/2}}{s_j - w_2} \right) \right. \\
 & + (z - \bar{z}) e^{i\beta_j} \frac{1}{s_j - \bar{w}_2} \left\{ - \left(\frac{e^{i\beta_j}}{zs_j} \right)^{1/2} + \frac{2}{s_j - \bar{w}_2} \left(-e^{i\beta_j} + \left(\frac{s_j e^{i\beta_j}}{z} \right)^{1/2} \right) \right\} \\
 & \left. + I_1(z, s_j) + I_2(z, s_j) \right] \\
 H_{j,0}(z, s_j) = & \frac{i}{16\pi} [I_1(z, s_j) - I_2(z, s_j)]
 \end{aligned} \quad (5.2.1)$$

For $m \geq 1$:

$$\begin{aligned}
 G_{j,m}(z, s_j) = & \frac{1}{16\pi} \sum_{k=0}^{m-1} \left\{ 2[\exp\{i(k - \tfrac{1}{2})\beta_j\} - i \sin(k + \tfrac{3}{2})\beta_j] A_{m-1,k} \right. \\
 & + i \sin \beta_j \exp\{i(k + \tfrac{1}{2})\beta_j\} B_{m-1,k} + (z - \bar{z}) [i \sin(k + \tfrac{3}{2})\beta_j \\
 & \times \exp(-i(k - \tfrac{1}{2})\beta_j) D_{m-1,k} + 2i \sin \beta_j \exp(-i(k + \tfrac{1}{2})\beta_j) E_{m-1,k}] \left. \right\} \\
 H_{j,m}(z, s_j) = & \frac{i}{16\pi} \sum_{k=0}^{m-1} \left\{ 2 \cos(k + \tfrac{3}{2})\beta_j \cdot A_{m-1,k} - i \sin \beta_j \exp(i(k + \tfrac{1}{2})\beta_j) B_{m-1,k} \right. \\
 & \left. + (z - \bar{z}) \{ -\cos(k + \tfrac{3}{2})\beta_j D_{m-1,k} + 2i \sin \beta_j \exp(-i(k + \tfrac{1}{2})\beta_j) E_{m-1,k} \} \right\}
 \end{aligned} \quad (5.2.2)$$

with the following expressions (5.3) and (5.4):

$$\begin{aligned}
 w_1 &= ze^{-i\beta_j}, \quad w_2 = \bar{z}e^{-i\beta_j} \\
 I_1(z, s_j) &= -e^{-i\beta_j} \left(\frac{1}{s_j - \bar{w}_1} - \frac{1}{s_j - \bar{w}_2} \right) - \left(\frac{e^{-i\beta_j}}{s_j} \right)^{1/2} \left(\frac{\bar{z}^{1/2}}{s_j - \bar{w}_1} + \frac{z^{1/2}}{s_j - \bar{w}_2} \right) \\
 &\quad - \cos \beta_j \left(\frac{e^{-i\beta_j}}{s_j} \right)^{1/2} \left(\frac{z^{1/2}}{s_j - w_1} + \frac{\bar{z}^{1/2}}{s_j - w_2} \right) + \frac{z - \bar{z}}{s_j - \bar{w}_2} \left[\frac{1}{2} \cos \beta_j \left(\frac{e^{i\beta_j}}{zs_j} \right)^{1/2} \right. \\
 &\quad \left. + \left\{ (e^{i\beta_j} - 2e^{-i\beta_j}) \left(\frac{s_j e^{i\beta_j}}{z} \right)^{1/2} + 1 \right\} \frac{1}{(s_j - \bar{w}_2)} \right. \\
 &\quad \left. + 4is_j \sin \beta_j \left\{ e^{i\beta_j} - \left(\frac{s_j e^{i\beta_j}}{z} \right)^{1/2} \right\} \frac{1}{(s_j - \bar{w}_2)^2} \right] \\
 I_2(z, s_j) &= e^{i\beta_j} \left(\frac{1}{s_j - w_1} - \frac{1}{s_j - w_2} \right) \\
 &\quad - 2i \sin \beta_j \left\{ s_j \left(\frac{1}{(s_j - w_1)^2} - \frac{1}{(s_j - w_2)^2} \right) + (s_j e^{-i\beta_j})^{1/2} \left(\frac{z^{1/2}}{(s_j - w_1)^2} + \frac{\bar{z}^{1/2}}{(s_j - w_2)^2} \right) \right\} \\
 &\quad + \frac{z - \bar{z}}{s_j - w_1} \left[-\frac{1}{2} \left(\frac{e^{i\beta_j}}{s_j z} \right)^{1/2} + \left\{ 1 + \left(\frac{s_j e^{i\beta_j}}{z} \right)^{1/2} \right\} \frac{1}{s_j - w_1} \right]
 \end{aligned} \tag{5.3}$$

$$\begin{aligned}
 A_{m,k} &= \frac{(-1)^m}{2^{3m+3}} K_{m,k} (z^{m-k+(1/2)} + \bar{z}^{m-k+(1/2)}) s_j^{k-(1/2)} \\
 B_{m,k} &= (1-2k) A_{m,k} (z^{m-k+(1/2)} + \bar{z}^{m-k+(1/2)}) s_j^{k-(1/2)} \\
 D_{m,k} &= \frac{(-1)^{m+1}}{2^{3m+3}} (2k+1) K_{m,m-k} z^{m-k-(3/2)} s_j^{k-(1/2)} \\
 E_{m,k} &= \frac{(-1)^m}{2^{3m+5}} [K_{m,k} - 2k(2k+1)K_{m,m-k} + 2(k+1)K_{m,m-k-1}] z^{m-k-(3/2)} s_j^{k-(1/2)} \\
 K_{m,k} &= \frac{1}{2m-2k-1} \binom{2k}{k} \binom{2m-2k}{m-k}
 \end{aligned} \tag{5.4}$$

2.3. Determination of force doublets distributed along branches

We proceed to the three loading cases as shown by Fig. 2. First, the isolated force doublets in (5.1) are replaced by continuous distributions with the following expressions [11,7]:

$$T_j^{[k]} = 4\sqrt{s_j(2b_j - s_j)} f_j^{[k]}(s_j) ds_j, \quad S_j^{[k]} = 4\sqrt{s_j(2b_j - s_j)} g_j^{[k]}(s_j) ds_j \quad (k = 1, 2, 3) \tag{6}$$

where the symbols $[k]$ denote the loading cases. The terms $[s_j(2b_j - s_j)]^{1/2}$ represent the stress singularity at the branch tip A_j , and $f_j^{[k]}(s_j)$, $g_j^{[k]}(s_j)$ are the weighting functions.

Now we use a perturbation procedure in order to get power series formulae of the stress intensity factors. For this purpose, we fix β_j and b_j ($j = 1, 2, \dots, N$), and take a as the variable. Then the weighting functions $f_j^{[k]}(s_j)$ and $g_j^{[k]}(s_j)$ in (6) are assumed as the following negative power series of a :

$$\left. \begin{aligned} \left\{ \begin{aligned} f_j^{[k]}(s_j) \\ g_j^{[k]}(s_j) \end{aligned} \right\} &= \sum_{n=0}^{\infty} \left\{ \begin{aligned} f_{j,n}^{[k]}(s_j) \\ g_{j,n}^{[k]}(s_j) \end{aligned} \right\} a^{-n_1} \\ n_1 &= \begin{cases} n & (k = 1, 3) \\ n + \frac{1}{2} & (k = 2) \end{cases} \end{aligned} \right\}. \quad (7)$$

It is to be noted that the terms in a^{-n} for cases [1] and [3] are replaced by $a^{-(n+(1/2))}$ for the case [2], according to (3.1) of the resultant forces due to the loads at infinity. New unknown functions $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$ ($k = 1, 2, 3; j = 1, 2, \dots, N; n = 0, 1, 2, \dots$) in the righthand sides of (7) are to be determined from the boundary conditions along the branches.

Substitute (6) and (7) into (5.1), integrate with respect to s_j , add for $j = 1$ to N and rearrange it in negative power series of a . Then the resultant forces at an arbitrary point z , due to the force doublets distributed along all the branches, are given as follows:

$$\left. \begin{aligned} R_y^{[k]} + iR_x^{[k]} &= 4 \sum_{m=0}^{\infty} a^{-m_1} \sum_{n=0}^m \sum_{j=1}^N \int_0^{2b_j} \sqrt{s_j(2b_j - s_j)} \\ &\quad \times [G_{j,m-n}(z, s_j) f_{j,n}^{[k]}(s_j) + H_{j,m-n}(z, s_j) g_{j,n}^{[k]}(s_j)] ds_j \\ m_1 &= \begin{cases} m & (k = 1, 3) \\ m + \frac{1}{2} & (k = 2) \end{cases} \end{aligned} \right\}. \quad (8)$$

The next step is to determine the unknown functions $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$ so that the resultant forces given by (8) eliminate those given by (3.1) due to the applied loads at infinity, at all the points on the branches. It cannot be done analytically, and a reliable numerical procedure is used. We divide the j th branch into M_j equal intervals, and assume $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$ by piecewise linear expressions as shown by Fig. 5. Then the unknown functions $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$ are uniquely defined by the independent unknowns $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$, which are the values at the

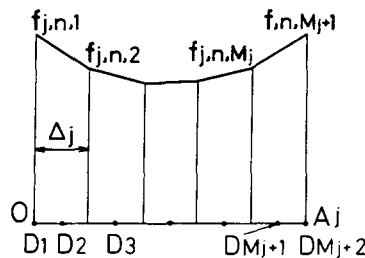


Fig. 5. Piecewise linear pattern of $f_{j,n}^{[k]}(s_j)$.

ends of the intervals. Actually we have the following expressions for $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$ in the p th interval of the j th branch:

$$\left. \begin{aligned} f_{j,n}^{[k]}(s_j) &= \frac{1}{\Delta_j} [(s_{j,p+1} - s_j) f_{j,n,p}^{[k]} + (s_j - s_{j,p}) f_{j,n,p+1}^{[k]}] \\ g_{j,n}^{[k]}(s_j) &= \frac{1}{\Delta_j} [(s_{j,p+1} - s_j) g_{j,n,p}^{[k]} + (s_j - s_{j,p}) g_{j,n,p+1}^{[k]}] \\ \Delta_j &= \text{interval length} = \frac{2b_j}{M_j} \\ (s_{j,p} \leq s_j \leq s_{j,p+1}; \quad p = 1, 2, \dots, M_j) \end{aligned} \right\} \quad (9)$$

where $s_{j,p}$ is the local coordinate of the left end of the p th interval of the j th branch.

Corresponding to this, we divide the integrals of (8) into the above defined M_j intervals and use the relations (9). Then the integrals in (8) can be evaluated analytically, and are reduced to the following linear form of the independent unknowns $f_{j,n}^{[k]}$ and $g_{j,n}^{[k]}$ ($j = 1, 2, \dots, N$; $p = 1, 2, \dots, M_j + 1$; $n = 0, 1, \dots$):

$$\left. \begin{aligned} R_y^{[k]} + iR_x^{[k]} &= \sum_{m=0}^{\infty} a^{-m_1} \sum_{n=0}^m \sum_{j=1}^N \sum_{p=1}^{M_j+1} [G_{j,m-n,p}(z) f_{j,n,p}^{[k]} + H_{j,m-n,p}(z) g_{j,n,p}^{[k]}] \\ m_1 &= \begin{cases} m & (k = 1, 3) \\ m + \frac{1}{2} & (k = 2) \end{cases} \end{aligned} \right\} \quad (10.1)$$

where

$$\left. \begin{aligned} G_{j,m-n,p}(z) &= \frac{4}{\Delta_j} \left[\int_{s_{j,p}}^{s_{j,p+1}} \sqrt{(s_j(2b_j - s_j))} G_{j,m-n}(z, s_j) (s_{j,p+1} - s_j) ds_j \right. \\ &\quad \left. + \int_{s_{j,p-1}}^{s_{j,p}} \sqrt{(s_j(2b_j - s_j))} G_{j,m-n}(z, s_j) (s_j - s_{j,p-1}) ds_j \right] \\ H_{j,m-n,p}(z) &= \frac{4}{\Delta_j} \left[\int_{s_{j,p}}^{s_{j,p+1}} \sqrt{(s_j(2b_j - s_j))} H_{j,m-n}(z, s_j) (s_{j,p+1} - s_j) ds_j \right. \\ &\quad \left. + \int_{s_{j,p-1}}^{s_{j,p}} \sqrt{(s_j(2b_j - s_j))} H_{j,m-n}(z, s_j) (s_j - s_{j,p-1}) ds_j \right] \end{aligned} \right\} \quad (10.2)$$

The integrals in (10.2) have been obtained in explicit forms, but they are not shown here to save the space.

The traction-free conditions along the branches require that the resultant force of each of the branch intervals due to the applied force doublets cancel the resultant force of the same interval caused by the stress at infinity. In order to get these relations in the same number as the unknowns, we consider new intervals $D_1 D_2, D_2 D_3, \dots, D_{M_t+1} D_{M_t+2}$ for the t th branch as shown by Fig. 5, where D_1 and D_{M_t+2} coincide with the both ends of the t th branch, and D_2 to D_{M_t+1}

are the midpoints of the previously defined equal intervals. Then the traction-free conditions along the branches can be replaced by the following relations in terms of the resultant forces:

$$\left. \begin{aligned} [R_y^{[k]} + iR_x^{[k]}]_{r_{t,q}}^{r_{t,q+1}} &= -[R_y^{*[k]} + iR_x^{*[k]}]_{r_{t,q}}^{r_{t,q+1}} \\ (t = 1, 2, \dots, N; \quad q = 1, 2, \dots, M_t + 1) \end{aligned} \right\} \quad (11)$$

where $r_{t,q}$ is the coordinate of the point D_q measured from the branching point along the t th branch.

Both sides of (11) are expressed as negative power series of a by (10.1) and (3.1). Equating the coefficients of the terms of the same powers in both sides, we get the following relations:

$$\left. \begin{aligned} & \sum_{n=0}^m \sum_{j=1}^N \sum_{p=1}^{M_j+1} (G_{j,m-n,p,t,q} f_{j,n,p}^{[k]} + H_{j,m-n,p,t,q} g_{j,n,p}^{[k]} = -\sigma \sqrt{\pi a} \dot{R}_{m,t,q}^{[k]} \\ & (k = 1, 2, 3; m = 0, 1, 2, \dots; t = 1, 2, \dots, N; q = 1, 2, \dots, M_t + 1) \end{aligned} \right\} \quad (12.1)$$

where

$$\left. \begin{aligned} G_{j,m-n,p,t,q} &= [G_{j,m-n,p}(z)]_{r_{t,q}}^{r_{t,q}+1}, \quad H_{j,m-n,p,t,q} = [H_{j,m-n,p}(z)]_{r_{t,q}}^{r_{t,q}+1} \\ \dot{R}_{m,t,q}^{[k]} &= [\dot{R}_{m,t,q}^{[k]}(z)]_{r_{t,q}}^{r_{t,q}+1} \end{aligned} \right\}, \quad (12.2)$$

Putting $m = 0, 1, 2, \dots$ in (12.1), we have

$$\sum_{j=1}^N \sum_{p=1}^{M_j+1} (G_{j,0,p,t,q} f_{j,0,p}^{[k]} + H_{j,0,p,t,q} g_{j,0,p}^{[k]}) = -\sigma \sqrt{\pi a} \dot{R}_{0,t,q}^{\star} \quad (13.1)$$

$$\sum_{j=1}^N \sum_{p=1}^{M_j+1} (G_{j,1,p,t,q} f_{j,0,p}^{[k]} + H_{j,1,p,t,q} g_{j,0,p}^{[k]} + G_{j,0,p,t,q} f_{j,1,p}^{[k]} + H_{j,0,p,t,q} g_{j,1,p}^{[k]}) = -\sigma \sqrt{\pi a} \dot{R}_{1,t,q}^{*[k]} \quad (13.2)$$

$$(k = 1, 2, 3; \quad t = 1, 2, \dots, N; \quad q = 1, 2, \dots, M_t + 1)$$

Each of these equations gives $\sum_{i=1}^N (2M_i + 2)$ relations, since $G_{j,l,p,t,q}$, $H_{j,l,p,t,q}$ and $\tilde{R}_{m,t,q}^{[k]}$ are complex constants, and the unknowns $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$ can be determined successively from the above equations. First we calculate $f_{j,0,p}^{[k]}$, $g_{j,0,p}^{[k]}$ ($j = 1, 2, \dots, N$; $p = 1, 2, \dots, M_{j+1}$) from (13.1), substitute these values in (13.2) and determine $f_{j,1,p}^{[k]}$ and $g_{j,1,p}^{[k]}$ ($j = 1, 2, \dots, N$; $p = 1, 2, \dots, M_{j+1}$), and so on. Thus we get $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$ for $n = 0, 1, \dots, L$ successively without solving simultaneous equations, where L is an arbitrarily assigned upper limit of n . Note that $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$ are determined exactly for any fixed subdivision numbers M_j ($j = 1, 2, \dots, N$), since the coefficients $G_{j,l,p,t,q}$ and $H_{j,l,p,t,q}$ in (13) have been evaluated analytically.

2.4. Stress intensity factors

Let $K_{I,A_j}^{[k]}$ and $K_{II,A_j}^{[k]}$ be the mode I and mode II stress intensity factors at the tip of the j th branch for the fundamental loading case $[k]$. They can be calculated from the obtained values of $f_{j,n,M_{j+1}}^{[k]}(s_j)$ and $g_{j,n,M_{j+1}}^{[k]}$ ($n = 0, 1, \dots, L$) by the following relations:

$$\left. \begin{aligned} K_{I,A_j}^{[k]} &= \sqrt{\pi b_j} \sum_{n=0}^L f_{j,n,M_{j+1}}^{[k]} a^{-n_1}, \quad K_{II,A_j}^{[k]} = \sqrt{\pi b_j} \sum_{n=0}^L g_{j,n,M_{j+1}}^{[k]} a^{-n_1} \\ n_1 &= \begin{cases} n & (k = 1, 3) \\ n + \frac{1}{2} & (k = 2) \end{cases} \end{aligned} \right\}. \quad (14)$$

Now we calculate $K_{I,B}$ and $K_{II,B}$ at the left tip of the main crack. Let $\Phi_j^{(T)}(z)$, $\Omega_j^{(T)}(z)$ and $\Phi_j^{(S)}(z)$, $\Omega_j^{(S)}(z)$ be the complex potentials for the isolated force doublets T_j and S_j shown by Fig. 4, (I) and (II), respectively. They can be derived by proper superposition of (4.1) with (4.2), and $K_{I,B}$, $K_{II,B}$ due to these force doublets are given as follows:

$$\begin{aligned} [K_I - iK_{II}]_B &= -2i\sqrt{2\pi} \lim_{z \rightarrow -2a} \sqrt{z + 2a} [\Phi_j^{(T)}(z) + \Phi_j^{(S)}(z)] \\ &= \sum_{m=0}^{\infty} a^{-(m+1)} [T_j U_{j,m}(s_j) + S_j V_{j,m}(s_j)] \end{aligned} \quad (15.1)$$

$$\left. \begin{aligned} U_{j,m} &= -\frac{1}{4} \sqrt{\frac{2}{\pi}} \frac{(-1)^{m+1}}{2^{3m+5}} \frac{(m+1)(m+2)}{2m+3} \left(\frac{2m+4}{m+2} \right) s_j^{m-(1/2)} \\ &\quad \times [2i \sin(m + \frac{3}{2})\beta_j + i(2m-1) \sin \beta_j \exp\{-i(m + \frac{1}{2})\beta_j\} + 2 \exp\{-i(m - \frac{1}{2})\beta_j\}] \\ V_{j,m} &= \frac{i}{4} \sqrt{\frac{2}{\pi}} \frac{(-1)^{m+1}}{2^{3m+5}} \frac{(m+1)(m+2)}{2m+3} \left(\frac{2m+4}{m+2} \right) s_j^{m-(1/2)} \\ &\quad \times [2 \cos(m + \frac{3}{2})\beta_j - i(2m-1) \sin \beta_j \exp\{-i(m + \frac{1}{2})\beta_j\}] \end{aligned} \right\}. \quad (15.2)$$

We substitute (6) for the three loading cases in (15.1), integrate with respect to s_j and write the righthand sides in terms of the body forces distributed along the j th branch. Summing up the resulting expressions for all j and expanding in negative power series of a with the aid of (7), we have the following forms:

$$\left. \begin{aligned} [K_I - iK_{II}]_B^{[k]} &= 4 \sum_{m=0}^L a^{-(m+1)} \sum_{n=0}^m \sum_{j=1}^N \int_0^{2b_j} \sqrt{s_j(2b_j - s_j)} \\ &\quad \times [U_{j,m-n}(s_j) f_{j,n}^{[k]}(s_j) + V_{j,m-n}(s_j) g_{j,n}^{[k]}(s_j)] ds_j \\ m_1 &= \begin{cases} m & (k = 1, 3) \\ m + \frac{1}{2} & (k = 2) \end{cases} \end{aligned} \right\}. \quad (16)$$

We divide the integrals in the righthand sides into the M_j intervals as shown by Fig. 5, and use the piecewise linear expressions of $f_{j,n}^{[k]}(s_j)$ and $g_{j,n}^{[k]}(s_j)$, Eqns. (9). Equations (16) are then reduced to linear forms in $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$, and the coefficients given by integrals are evaluated analytically. Using the values of $f_{j,n,p}^{[k]}$ and $g_{j,n,p}^{[k]}$ determined by (13), the righthand sides of (16) are obtained in forms of power series. Combining these results with the stress intensity factors of the line crack BO under the applied stresses at infinity, we get the final results of $[K_I - iK_{II}]_B^{[k]}$ at the left tip B of the main crack.

In representing the above obtained results of $K_I^{[k]}$ and $K_{II}^{[k]}$ at the crack tips A_j ($j = 1, 2, \dots, N$) and B, we define the following dimensionless quantities:

$$\left. \begin{aligned} F_{I,A_j}^{[k]} &= \frac{K_{I,A_j}^{[k]}}{\sigma\sqrt{\pi a}}, & F_{II,A_j}^{[k]} &= \frac{K_{II,A_j}^{[k]}}{\sigma\sqrt{\pi a}}, \\ F_{I,B}^{[k]} &= \frac{K_{I,B}^{[k]}}{\sigma\sqrt{\pi a}}, & F_{II,B}^{[k]} &= \frac{K_{II,B}^{[k]}}{\sigma\sqrt{\pi a}} \end{aligned} \right\}. \quad (17)$$

$(k = 1, 2, 3; \quad j = 1, 2, \dots, N)$

3. Numerical results

3.1. Power series formulae and accuracy of the results

We finally get power series expressions of the dimensionless stress intensity factors for each of the assigned branch patterns, where the branch pattern includes N , β_j and the ratios of b_j . The obtained formulae are as follows:

$$\left. \begin{aligned} \left\{ \begin{array}{c} F_{I,A_j}^{[k]} \\ F_{II,A_j}^{[k]} \\ F_{I,B}^{[k]} \\ F_{II,B}^{[k]} \end{array} \right\} &= \sum_{n=0}^L \left\{ \begin{array}{c} F_{I,A_j,n}^{[k]} \\ F_{II,A_j,n}^{[k]} \\ F_{I,B,n}^{[k]} \\ F_{II,B,n}^{[k]} \end{array} \right\} \lambda^n, & \left\{ \begin{array}{c} F_{I,A_j}^{[2]} \\ F_{II,A_j}^{[2]} \\ F_{I,B}^{[2]} \\ F_{II,B}^{[2]} \end{array} \right\} &= \sum_{n=0}^L \left\{ \begin{array}{c} F_{I,A_j,n}^{[2]} \\ F_{II,A_j,n}^{[2]} \\ F_{I,B,n}^{[2]} \\ F_{II,B,n}^{[2]} \end{array} \right\} \lambda^{n+(1/2)} \end{aligned} \right\}, \quad (18)$$

$(k = 1, 3; \quad j = 1, 2, \dots, N)$

$\lambda = \frac{b}{a}$

where b is the half length of one of the branches, and we may take $b = b_1$ arbitrarily.

Note that the first terms in (18) corresponding to $n = 0$ give $F_{I,A_j}^{[k]}$, $F_{II,A_j}^{[k]}$, $F_{I,B}^{[k]}$ and $F_{II,B}^{[k]}$ for cracks with extremely short branches. The coefficients in the righthand sides of (18) can be calculated successively for as many as required by using (13), (14) and (16). In the current work, the upper limit L of the summations in (18) is taken to be 20.

Here some comments will be made on the accuracy of the obtained coefficients of (18). As mentioned in Section 2.3, $f_{j,n,M_j+1}^{[k]}$ and $g_{j,n,M_j+1}^{[k]}$ in (14) have been determined exactly from (13) for any assigned values of M_j . Therefore, the numerical errors of $F_{I,A_j,n}^{[k]}$ and $F_{II,A_j,n}^{[k]}$ in (18) depend only upon the subdivision numbers M_j , and can be estimated from the convergence as M_j increase.

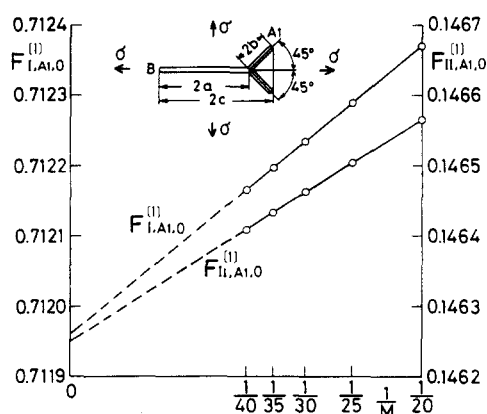


Fig. 6. $F_{I,A1,0}^{(k)}$ and $F_{II,A1,0}^{(k)}$ vs $1/M$ for symmetrically branched crack with opening angle of 90° .

Figure 6 shows the examples of $F_{I,A1,n}^{[1]}$ and $F_{II,A1,n}^{[1]}$ for symmetrically branched cracks with the opening angle of 90° subjected to all-around tension, which give $F_{I,A1}$ and $F_{II,A1}$ for the crack with extremely short branches. The values for various subdivisions are plotted against $1/M$, where M is the number of subdivision, and they are shown to be nearly linear with $1/M$. Therefore, the exact values corresponding to $M \rightarrow \infty$ can be estimated by linear extrapolation of the values for $M = 35$ and 40 as shown by dashed lines. Apart from this, even the raw values for $M = 20$ are very accurate. The estimated errors seem less than 0.001 , as is seen from magnified scales of the ordinates, left for $F_{I,A1,0}^{[1]}$ and right for $F_{II,A1,0}^{[1]}$.

3.2. Formulae for cracks with two symmetrical branches

3.2.1. Power series formulae for $F_I^{(k)}$ and $F_{II}^{(k)}$

We consider the cracks with two symmetrical branches ($N = 2$, $b_1 = b_2 = b$ and $\beta_1 = -\beta_2 = \beta$). Computations have been performed in each of the fundamental cases, [1], [2] and [3] for 13 values of β as is seen in Table 2, and the series formulae (18) are obtained up to the terms in $(b/a)^{20}$ for the cases [1] and [3], and $(b/a)^{20+1/2}$ for the case [2]. The coefficients in (18) have been determined by linear extrapolation as shown in Fig. 6, but we assumed $M = 30$ and 35 as two subdivision numbers differently from Fig. 6 in order to save the computing cost, since they are enough to give reliable values accurate up to the third decimal places. The convergence of the obtained series (18) is considered to be fair for all the calculated cases from general trends of the coefficients of (18). The numerical results from (18) with $L = 20$ will be discussed in Section 3.3.

For cracks with short branches, most of the previous analyses give unreliable results, but the series formulae (18) give highly accurate values with only the first few terms. To get a general idea on the relation between b/a and the value of n in (18) at which the series can be truncated with sufficient accuracy, we consider 90° -branched cracks ($b_1 = b_2 = b$, $\beta_1 = -\beta_2 = 45^\circ$) subjected to uniaxial tension vertical to the main crack which is realized by superposing the cases [1] and [2]. This is a special case of Problem (a) in Fig. 7 and to be discussed in Figs. 8 to 10 and Fig. 13 of Section 3.3. Table 1 shows the results of $F_I' = K_I/\sigma\sqrt{\pi c}$ and $F_{II}' = K_{II}/\sigma\sqrt{\pi c}$ at crack tips A_1 and B for various values of b/a , where $c = a + b \cos \beta$ is half the projected length of the total crack in the direction of the main crack. In the table are also given the previous values by Isida [7] and by Kitagawa et al. [4]. These results agree well with each other in the

Table 1. Values for the case of $\beta = 45^\circ$ in Problem (a) of Fig. 7, where $F'_I = K_{II}/\sigma\sqrt{\pi c}$ and $F'_{II} = K_{II}/\sigma\sqrt{\pi c}$

$\frac{b}{a}$	n (1%)	n (0.1%)	F'_{I,A_1}		F'_{II,A_1}		$F'_{L,B}$	
			Authors	Ref. [7]	Ref. [4]	Authors	Ref. [7]	Ref. [4]
0	0	0	0.712			1.000		
0.001	0	0	0.692			1.000		
0.005	0	1	0.669			1.001		
0.01	0	1	0.652			1.002		
0.02	1	1	0.631			1.003	1.003	
0.03	1	1	0.615	0.624		1.004	1.004	
0.05	1	1	0.593	0.593		1.006	1.006	1.01
0.1	1	2	0.560	0.560	0.59	1.010	1.010	1.01
0.2	2	3	0.528	0.528	0.56	1.015	1.015	1.02
0.3	2	4	0.512	0.512	0.54	1.019	1.019	
0.4	3	5	0.504	0.504	0.51	1.023	1.023	1.03
0.5	5	10	0.500	0.500		1.026	1.026	
0.6	7	15	0.497	0.497	0.50	1.029	1.029	1.04
0.7	11	17	0.496	0.496		1.032	1.033	

Note: Values in parentheses show the truncation errors.

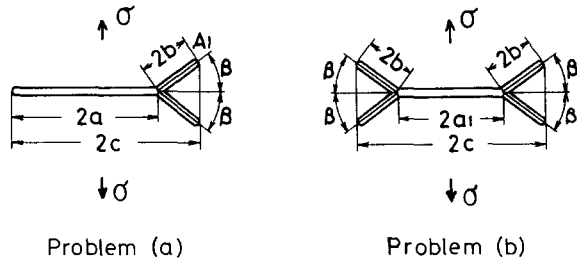


Fig. 7. Cracks with two symmetrical branches at either or both ends.

Table 2. Coefficients for $n \leq 2$ in Eqns. (18) for cracks with two symmetrical branches

$\beta(^{\circ})$	$F_{I,A_1,0}^{[1]}$	$F_{I,A_1,1}^{[1]}$	$F_{I,A_1,2}^{[1]}$	$F_{II,A_1,0}^{[1]}$	$F_{II,A_1,1}^{[1]}$	$F_{II,A_1,2}^{[1]}$	$F_{I,B,0}^{[1]}$	$F_{I,B,1}^{[1]}$	$F_{I,B,2}^{[1]}$
2.5	0.689	0.346	-0.0873	-0.195	-0.1224	0.0369	1.0000	0.5097	-0.1304
5	0.6986	0.3548	-0.0916	-0.1779	-0.1339	0.0475	1.0000	0.5189	-0.1352
10	0.7178	0.3799	-0.1062	-0.1409	-0.1742	0.0659	1.0000	0.5345	-0.1435
15	0.7325	0.4129	-0.1269	-0.1015	-0.2061	0.0771	1.0000	0.5462	-0.1499
20	0.7422	0.4531	-0.1508	-0.0601	-0.2336	0.0779	1.0000	0.5539	-0.1541
30	0.7460	0.5500	-0.1941	0.0246	-0.2680	0.0412	1.0000	0.5565	-0.1555
40	0.7285	0.6574	-0.2049	0.1074	-0.2668	-0.0447	1.0000	0.5417	-0.1472
45	0.7119	0.7100	-0.1894	0.1463	-0.2513	-0.1006	1.0000	0.5278	-0.1398
50	0.6906	0.7589	-0.1572	0.1826	-0.2260	-0.1600	1.0000	0.5099	-0.1304
60	0.6346	0.8371	-0.0418	0.2452	-0.1492	-0.2708	1.0000	0.4625	-0.1072
70	0.5640	0.8776	0.1287	0.2908	-0.0471	-0.3410	1.0000	0.4024	-0.0811
80	0.4833	0.8711	0.3224	0.3165	0.0645	-0.3455	1.0000	0.3335	-0.0558
90	0.3977	0.8158	0.4968	0.3211	0.1678	-0.2786	1.0000	0.2611	-0.0344
$\beta(^{\circ})$	$F_{I,A_1,0}^{[2]}$	$F_{I,A_1,1}^{[2]}$	$F_{I,A_1,2}^{[2]}$	$F_{II,A_1,0}^{[2]}$	$F_{II,A_1,1}^{[2]}$	$F_{II,A_1,2}^{[2]}$	$F_{I,B,0}^{[2]}$	$F_{I,B,1}^{[2]}$	$F_{I,B,2}^{[2]}$
2.5	0.0291	0.0011	-0.0012	0.1279	-0.0003	0.0004	0.0000	-0.0033	0.0035
5	0.0305	0.0032	-0.0035	0.1966	-0.0008	0.0010	0.0000	-0.0092	0.0104
10	0.0080	0.0092	-0.0106	0.3140	-0.0018	0.0027	0.0000	-0.0258	0.0316
15	-0.0392	0.0174	-0.0209	0.4181	-0.0024	0.0047	0.0000	-0.0475	0.0616
20	-0.1066	0.0272	-0.0343	0.5109	-0.0022	0.0064	0.0000	-0.0732	0.0995
30	-0.2904	0.0498	-0.0648	0.6570	0.0016	0.0073	0.0000	-0.1336	0.1938
40	-0.5201	0.0727	-0.1076	0.7398	0.0107	0.0023	0.0000	-0.1995	0.3022
45	-0.6443	0.0827	-0.1266	0.7543	0.0170	-0.0029	0.0000	-0.2321	0.3569
50	-0.7703	0.0909	-0.1438	0.7502	0.0240	-0.0099	0.0000	-0.2632	0.4091
60	-1.0143	0.1005	-0.1685	0.6870	0.0388	-0.0277	0.0000	-0.3166	0.4979
70	-1.2260	0.0995	-0.1756	0.5577	0.0513	-0.0465	0.0000	-0.3527	0.5545
80	-1.3834	0.0884	-0.1633	0.3778	0.0579	-0.0606	0.0000	-0.3658	0.5700
90	-1.4709	0.0704	-0.1353	0.1691	0.0568	-0.0658	0.0000	-0.3536	0.5427
$\beta(^{\circ})$	$F_{I,A_1,0}^{[3]}$	$F_{I,A_1,1}^{[3]}$	$F_{I,A_1,2}^{[3]}$	$F_{II,A_1,0}^{[3]}$	$F_{II,A_1,1}^{[3]}$	$F_{II,A_1,2}^{[3]}$	$F_{I,B,0}^{[3]}$	$F_{I,B,1}^{[3]}$	$F_{I,B,2}^{[3]}$
10	-0.726	-0.448	0.092	0.468	0.099	0.033	1.0000	0.711	-0.242
15	-0.8050	-0.4710	0.0467	0.4383	-0.0047	0.1039	1.0000	0.8274	-0.3363
20	-0.8675	-0.4653	-0.0298	0.4241	-0.1363	0.1825	1.0000	0.9264	-0.4253
30	-0.9778	-0.3489	-0.2801	0.3776	-0.4465	0.3186	1.0000	1.0966	-0.5993
40	-1.0686	-0.0871	-0.6205	0.3055	-0.7667	0.3623	1.0000	1.2339	-0.7601
45	-1.1055	0.0944	-0.7992	0.2613	-0.9110	0.3325	1.0000	1.2893	-0.8302
50	-1.1363	0.3044	-0.9667	0.2127	-1.0353	0.2647	1.0000	1.3352	-0.8906
60	-1.1775	0.7867	-1.2181	0.1055	-1.2002	0.0247	1.0000	1.3965	-0.9745
70	-1.1901	1.3030	-1.2900	-0.0091	-1.2275	-0.3073	1.0000	1.4152	-1.0009
80	-1.1731	1.7879	-1.1430	-0.1238	-1.1072	-0.6449	1.0000	1.3902	-0.9660
90	-1.1272	2.1781	-0.7973	-0.2312	-0.8552	-0.8941	1.0000	1.3229	-0.8748

Note: $F_{I,A_2,n}^{[k]} = F_{I,A_1,n}^{[k]}$, $F_{II,A_2,n}^{[k]} = -F_{II,A_1,n}^{[k]}$ ($k = 1, 2$) $F_{I,A_2,n}^{[3]} = -F_{I,A_1,n}^{[3]}$, $F_{II,A_2,n}^{[3]} = F_{II,A_1,n}^{[3]}$

range of $b/a \geq 0.05$, and the best results are obviously given by the present analysis especially for small values of b/a .

In Table 1 are also given the values of n at which the series can be truncated when 1 percent and 0.1 percent errors are allowed. We find that the series (18) can be truncated at $n = 1, 2, 5$ and 11 when $b/a = 0.1, 0.3, 0.5$ and 0.7 at the 1 percent error basis, respectively.

3.2.2. Formulate for $b/a \leq 0.3$

(i) One parametric power series.

We pay attention to cracks with short branches for which most of the previous analyses do not give reliable results. Here we consider the range $b/a \leq 0.3$, and in this range, the series (18) can be truncated at $n = 2$ as shown by Table 1.

The coefficients $F_{I,A_1,n}^{[k]}$, $F_{II,A_1,n}^{[k]}$, $F_{I,B,n}^{[k]}$ and $F_{II,B,n}^{[k]}$ ($k = 1, 2, 3$; $n = 0, 1, 2$) of (18) required in calculating the stress intensity factors at the tips A_1 and B for $b/a \leq 0.3$ are given in Table 2 for 13 values of β between 2.5° and 90° . The values for another tip A_2 are given by the following relations:

$$\left. \begin{aligned} F_{I,A_2,n}^{[1]} &= F_{I,A_1,n}^{[1]}, & F_{I,A_2,n}^{[2]} &= F_{I,A_1,n}^{[2]}, & F_{I,A_2,n}^{[3]} &= -F_{I,A_1,n}^{[3]} \\ F_{II,A_2,n}^{[1]} &= -F_{II,A_1,n}^{[1]}, & F_{II,A_2,n}^{[2]} &= -F_{II,A_1,n}^{[2]}, & F_{II,A_2,n}^{[3]} &= F_{II,A_1,n}^{[3]} \end{aligned} \right\} \quad (19)$$

(ii) Two parametric power series

We have proposed (18) in forms of power series with respect to b/a for each of the assigned values of β . But it is more desirable to give some formulae including β and b/a as explicit variables. With these formulae, we can get reliable values for arbitrary combinations of β and b/a without interpolating procedure which may cause considerable errors.

For this purpose, the analytical values of $F_{I,A_1,n}^{[k]}$ and $F_{II,A_1,n}^{[k]}$ ($n \leq 2$) for 18 values of β between 5° and 90° have been fitted to power series of β with a least square procedure, and the following polynomials are obtained:

$$F_{I,A_1,n}^{[k]}(\beta) = \sum_{p=0}^5 C_{I,n,p}^{[k]} \beta^p, \quad F_{II,A_1,n}^{[k]} = \sum_{p=0}^5 C_{II,n,p}^{[k]} \beta^p \quad (n \leq 2). \quad (20)$$

Substituting (20) into (18) truncated at $n = 2$, we finally get the following formulae of $F_{I,A_1}^{[k]}(\beta, \lambda)$ and $F_{II,A_1}^{[k]}(\beta, \lambda)$ for the three fundamental loads:

$$\begin{aligned} F_{I,A_1}^{[1]}(\beta, \lambda) &= 0.6752 + 0.5052\omega - 0.7087\omega^2 - 0.8759\omega^3 + 0.8640\omega^4 - 0.1459\omega^5 \\ &\quad + \lambda(0.3376 + 0.2664\omega + 1.4974\omega^2 + 1.1836\omega^3 - 4.9322\omega^4 + 2.3687\omega^5) \\ &\quad + \lambda^2(-0.0951 + 0.2354\omega - 4.0819\omega^2 + 7.6655\omega^3 - 0.5692\omega^4 - 2.5891\omega^5), \end{aligned} \quad (21.1)$$

$$\begin{aligned} F_{II,A_1}^{[1]}(\beta, \lambda) &= -0.2119 + 0.6452\omega + 0.7365\omega^2 - 0.7722\omega^3 - 0.4094\omega^4 + 0.3172\omega^5 \\ &\quad + \lambda(-0.1079 - 0.6026\omega - 1.0890\omega^2 + 5.5907\omega^3 - 4.2283\omega^4 + 0.6751\omega^5) \\ &\quad + \lambda^2(0.0332 + 0.2062\omega + 2.5354\omega^2 - 16.171\omega^3 + 21.679\omega^4 - 8.4433\omega^5), \end{aligned} \quad (21.2)$$

$$\begin{aligned}
F_{I,A_1}^{[3]}(\beta, \lambda) = & -0.5358 - 2.2716\omega + 4.2203\omega^2 - 7.3504\omega^3 + 7.9209\omega^4 - 3.0551\omega^5 \\
& + \lambda(-0.3481 - 1.1279\omega - 0.4614\omega^2 + 19.971\omega^3 - 21.632\omega^4 + 6.0104\omega^5) \\
& + \lambda^2(0.1118 - 0.0385\omega + 0.9145\omega^2 - 30.966\omega^3 + 53.075\omega^4 - 23.478\omega^5),
\end{aligned} \quad (21.3)$$

$$\begin{aligned}
F_{II,A_1}^{[3]}(\beta, \lambda) = & 0.5447 - 1.1666\omega + 5.0016\omega^2 - 13.753\omega^3 + 14.048\omega^4 - 5.0280\omega^5 \\
& + \lambda(0.1954 - 0.0690\omega - 9.8089\omega^2 + 8.3503\omega^3 + 5.9942\omega^4 - 5.1881\omega^5) \\
& + \lambda^2(0.0268 - 1.2590\omega + 17.078\omega^2 - 39.787\omega^3 + 27.176\omega^4 - 4.1425\omega^5).
\end{aligned} \quad (21.4)$$

Equations (21.1) to (21.4) are valid in the range $b/a \leq 0.3$, $5^\circ \leq \beta \leq 90^\circ$ with less than 1 percent errors.

$$\begin{aligned}
F_{I,A_1}^{[2]}(\beta, \lambda) = & \lambda^{1/2}(0.0296 + 0.2885\omega - 5.3443\omega^2 + 2.0060\omega^3 + 2.9697\omega^4 - 1.4258\omega^5) \\
& + \lambda^{3/2}(-0.0002 + 0.0418\omega + 0.5530\omega^2 - 0.2050\omega^3 - 1.0220\omega^4 + 0.6859\omega^5) \\
& + \lambda^{5/2}(0.0012 - 0.0717\omega - 0.3434\omega^2 - 1.5881\omega^3 + 3.7124\omega^4 - 1.8151\omega^5),
\end{aligned} \quad (21.5)$$

$$\begin{aligned}
F_{II,A_1}^{[2]}(\beta, \lambda) = & \lambda^{1/2}(0.0738 + 2.5413\omega - 1.2922\omega^2 - 2.2903\omega^3 - 0.4673\omega^4 + 1.4079\omega^5) \\
& + \lambda^{3/2}(-0.0008 + 0.0138\omega - 0.3745\omega^2 + 1.7190\omega^3 - 1.9914\omega^4 + 0.6807\omega^5) \\
& + \lambda^{5/2}(0.0020 - 0.0456\omega + 0.6958\omega^2 - 2.1116\omega^3 + 1.8174\omega^4 - 0.4120\omega^5).
\end{aligned} \quad (21.6)$$

Equations (21.5) and (21.6) are valid in the range $b/a \leq 0.3$, $10^\circ \leq \beta \leq 90^\circ$ with less than 1 percent errors.

The parameters in (21.1) to (21.6) are

$$\lambda = \frac{b}{a}, \quad \omega = \frac{\beta}{100} \quad (\beta: \text{in deg.}) \quad (21.7)$$

and the values for the other tip A_2 are given by (19).

We have proposed power series formulae for the three fundamental loads [1], [2] and [3] as shown by Fig. 2. In more general cases when the plate is subjected to $\sigma_x = p_x\sigma$, $\sigma_y = p_y\sigma$ and $\tau_{xy} = p_{xy}\sigma$ at infinity, F_I and F_{II} at all the tips are given by superposing the above results for fundamental loads as follows:

$$\left. \begin{aligned}
F_{I,A_1} &= \frac{K_{I,A_1}}{\sigma\sqrt{\pi a}} = p_y F_{I,A_1}^{[1]} + (p_y - p_x) F_{I,A_1}^{[2]} + p_{xy} F_{I,A_1}^{[3]} \\
F_{I,A_2} &= \frac{K_{I,A_2}}{\sigma\sqrt{\pi a}} = p_y F_{I,A_1}^{[1]} + (p_y - p_x) F_{I,A_1}^{[2]} - p_{xy} F_{I,A_1}^{[3]} \\
F_{I,B} &= \frac{K_{I,B}}{\sigma\sqrt{\pi a}} = p_y F_{I,B}^{[1]} + (p_y - p_x) F_{I,B}^{[2]}
\end{aligned} \right\}, \quad (22.1)$$

$$\left. \begin{aligned} F_{II,A_1} &= \frac{K_{II,A_1}}{\sigma\sqrt{\pi a}} = p_y F_{II,A_1}^{[1]} + (p_y - p_x) F_{II,A_1}^{[2]} + p_{xy} F_{II,A_1}^{[3]} \\ F_{II,A_2} &= \frac{K_{II,A_2}}{\sigma\sqrt{\pi a}} = -p_y F_{II,A_1}^{[1]} - (p_y - p_x) F_{II,A_1}^{[2]} + p_{xy} F_{II,A_1}^{[3]} \\ F_{II,B} &= \frac{K_{II,B}}{\sigma\sqrt{\pi a}} = p_{xy} F_{II,B}^{[3]} \end{aligned} \right\}, \quad (22.2)$$

where $F_{II,A_1}^{[1]}$ etc. are defined by (18) with Table 2 for various values of β and (21) and (19) are useful for arbitrary values of β and λ . The results given by (22) are valid as long as all of F_{I,A_1} , F_{I,A_2} and $F_{I,B}$ are non-negative.

As an example, we consider an arbitrary uniaxial tension of infinitesimal symmetrical branches as shown in Fig. 8. In this case, the coefficients p_x , p_y and p_{xy} in (22.1) are given by $(1 - \cos 2\alpha)/2$, $(1 + \cos 2\alpha)/2$ and $\sin 2\alpha/2$, respectively. For infinitesimal branches, $F_{I,A_1}^{[2]}$ and $F_{I,B}^{[2]}$ vanish and $F_{I,A_1}^{[1]}$, $F_{I,A_1}^{[3]}$, $F_{I,B}^{[1]}$ can be replaced by $F_{I,A_1,0}^{[1]}$, $F_{I,A_1,0}^{[3]}$, $F_{I,B,0}^{[1]}$, respectively. They are numerically given in Table 2 for various values of β , and we can determine the range for which non-negative values of F_{I,A_1} , F_{I,A_2} , and $F_{I,B}$ are obtained. Thus determined range of valid analysis is shown by the hatched areas in Fig. 8.

3.3. Numerical results for cracks with two symmetrical branches

In Sections 3.3 and 3.4, we will discuss typical numerical results based on (18) with $L=20$.

We are especially interested in cracks with two symmetrical branches ($N=2$, $\beta_1 = -\beta_2 = \beta$ and $b_1 = b_2 = b$), subjected to uniaxial tension which is realized by superposing the loading cases [1] and [2]. To begin with, let's compare the authors' results for this problem with previous values by Vitek for cracks branched at both the ends [5]. These problems (a) and (b) are shown by Fig. 7, and the dimensionless factors F'_{I,A_1} and F'_{II,A_1} are defined on the basis of $\sigma\sqrt{\pi c}$, where c is half the projected length of the total crack. Table 3 shows the authors' results

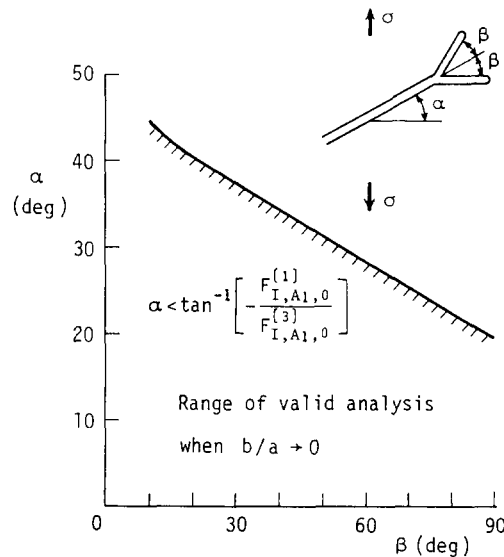


Fig. 8. Range of valid analysis for a crack with infinitesimal symmetrical branches under arbitrary uniaxial tension.

Table 3. Comparison of the results for Problems (a) and (b), where $F'_{I,A_1} = K_{I,A_1}/\sigma\sqrt{\pi c}$ and $F'_{II,A_1} = K_{II,A_1}/\sigma\sqrt{\pi c}$

β	$\frac{b}{c}$	F'_{I,A_1}		F'_{II,A_1}	
		Prob. (a) Authors	Prob. (b) Vitek [5]	Prob. (a) Authors	Prob. (b) Vitek [5]
5°	0.005	0.701	0.748	-0.164	-0.218
	0.045	0.705	0.755	-0.138	-0.194
	0.083	0.708	0.759	-0.125	-0.182
	0.167	0.712	0.766	-0.106	-0.165
	0.250	0.716	0.772	-0.092	-0.154
20°	0.005	0.735	0.734	-0.025	-0.021
	0.046	0.724	0.728	0.040	0.043
	0.084	0.721	0.726	0.071	0.073
	0.170	0.717	0.727	0.115	0.117
	0.258	0.717	0.732	0.145	0.146
45°	0.005	0.669	0.660	0.198	0.199
	0.047	0.595	0.590	0.295	0.297
	0.088	0.563	0.562	0.342	0.344
	0.185	0.525	0.529	0.410	0.413
	0.293	0.506	0.517	0.454	0.455

for Problem (a) calculated from (18) in comparison with Vitek's values for Problem (b) [5]. When b/c is small, the effects of local geometries near the left ends of the cracks will be negligible, and the authors' results are expected to be close to Vitek's. Actually we find good agreement between both the results when $\beta = 20^\circ$ and 45° especially for small values of b/c . In the case of $\beta = 5^\circ$, however, considerable differences are observed, and the authors' values are regarded as more reliable for small values of b/c , since the coefficients in (18) have been determined accurately by linear extrapolation as shown in Fig. 6.

Figures 9 to 13 give various results for cracks with two symmetrical branches, where logarithmic scales are used for the abscissa b/a , putting emphasis on the stress intensity factors for small values of b/a . Figures 9 and 10 show the variations of F_{I,A_1} and F_{II,A_1} with b/a and β for the uniaxial tension, in which F_{I,A_1} and F_{II,A_1} are defined on the basis of $\sigma\sqrt{\pi a}$. Figure 11 is a

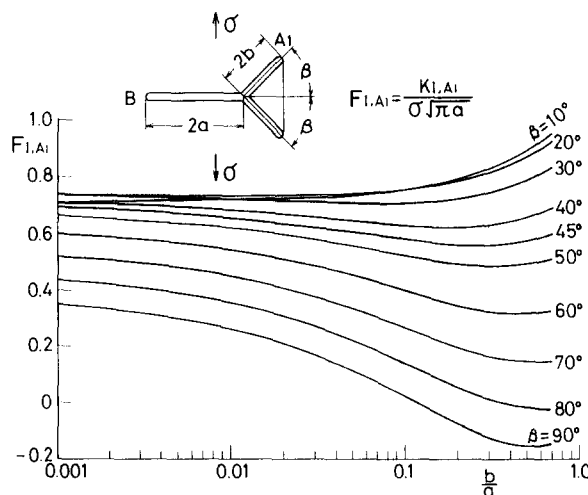


Fig. 9. F_{I,A_1} for symmetrically branched crack subjected to tension.

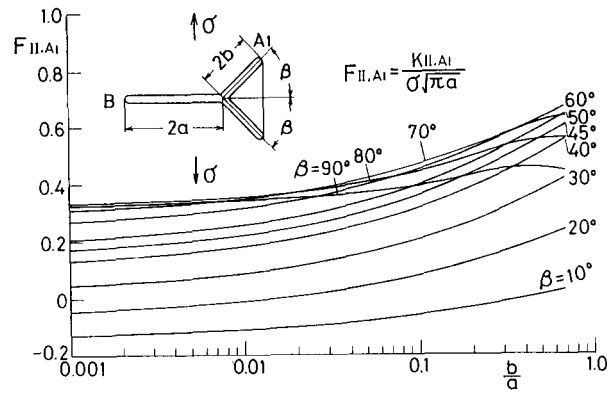


Fig. 10. $F_{II,A1}$ for symmetrically branched crack subjected to tension.

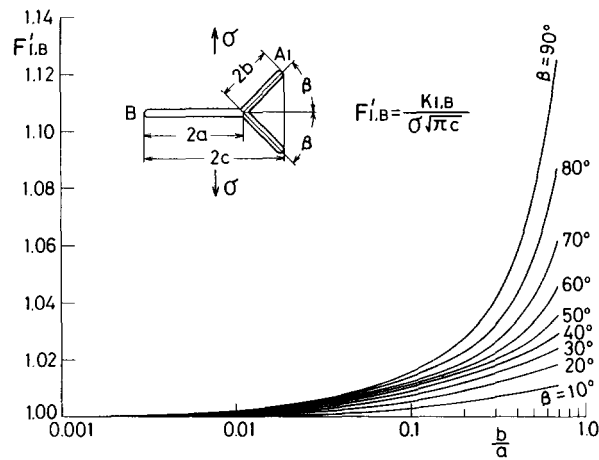


Fig. 11. $F'_{I,B}$ for symmetrically branched crack subjected to tension where $F'_{I,B} = K_{I,B} / \sigma \sqrt{\pi c}$, c = half the projected length.

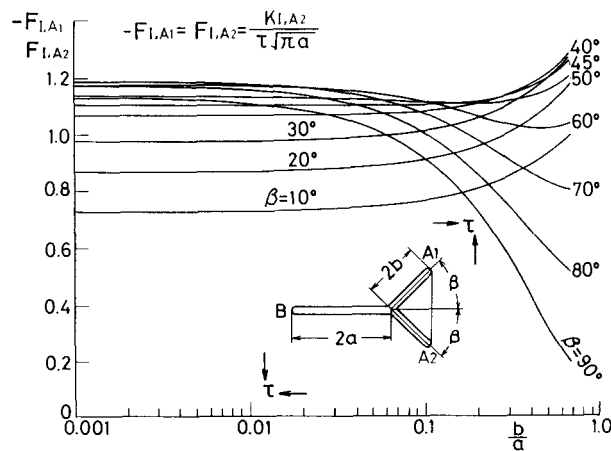


Fig. 12. $F_{I,A1}$ for symmetrically branched crack subjected to shear.

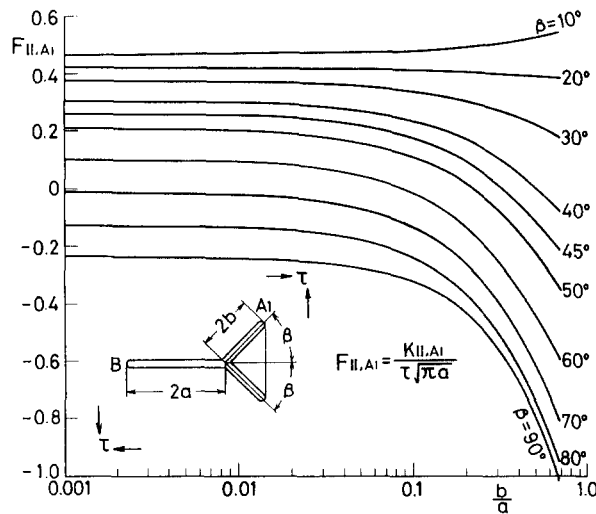


Fig. 13. $F_{II,A1}$ for symmetrically branched crack subjected to shear.

similar plot of $F'_{I,B}$, for the left tip of the main crack in the same problem, where $F'_{I,B} = K_{I,B}/\sigma\sqrt{\pi c}$ and c is half the projected length of the total crack. We find that $F'_{I,B}$ is very close to unity in a wide range of b/a and β , or $K_{I,B}$ is nearly equal to that for the line crack with length $2c$. Figures 12 and 13 show the variations of F_I and F_{II} at the tips A_1 and A_2 with b/a and β for the uniform shear. In this case, $F_{I,A1}$ is negative and the edges of the branch OA_1 contact, acting like a slip line instead of a free crack. Therefore the results in these figures are not realistic for this specific problem, but they may be used in getting valid solutions for some combined loads by superposition.

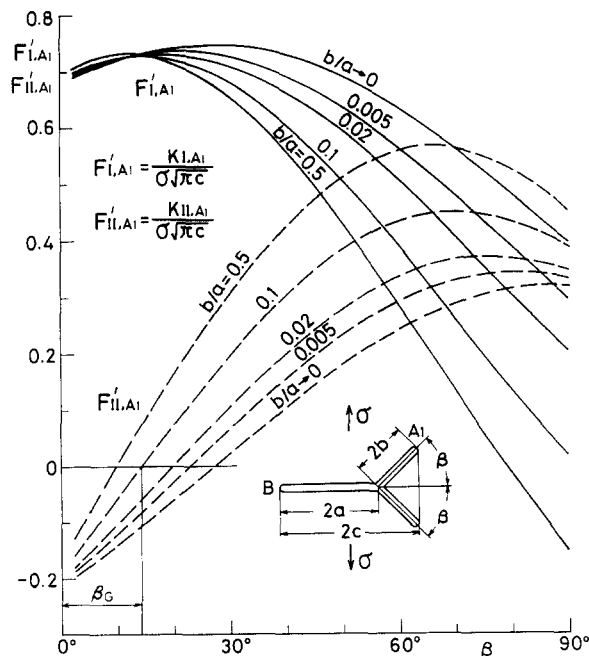
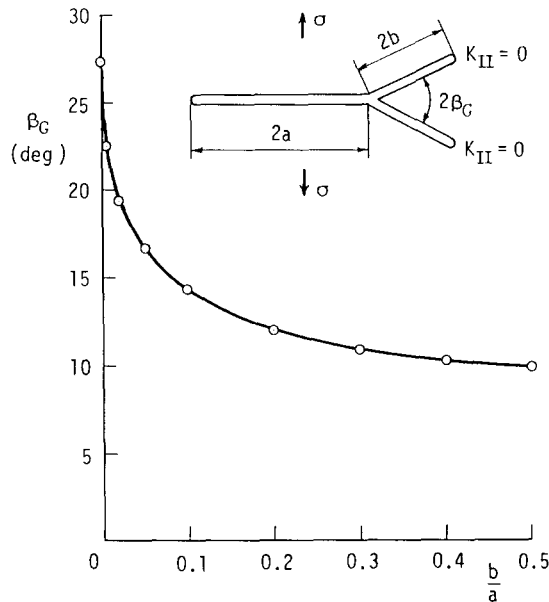


Fig. 14. $F'_{I,A1}$ and $F'_{II,A1}$ for symmetrically branched crack under tension, where $F'_{I,A} = K_{I,A}/\sigma\sqrt{\pi c}$, $F'_{II,A} = K_{II,A}/\sigma\sqrt{\pi c}$.

Table 4. Relation between b/a and β_G

b/a	0	0.005	0.02	0.05	0.1	0.2	0.3	0.4	0.5
β_G (deg.)	27.3	22.5	19.3	16.7	14.3	12.0	10.8	10.2	9.8

Fig. 15. Characteristic angle β_G vs b/a for symmetrically branched crack under tension.

The results of Figs. 9 and 10 for the uniaxial tension are redrawn in Fig. 14 in a more practical manner. The dimensionless factors F'_{I,A_1} and F'_{II,A_1} , based on $\sigma\sqrt{\pi c}$ with c as half the projected length of the total crack, are plotted against the branching angle β for various values of b/a . We find that, for any fixed values of b/a , F'_{II,A_1} vanishes at some value $\beta = \beta_G$ as shown by dashed curves, and at this value of β , F'_{I,A_1} takes its maximum approximately as shown by solid curves. This angle β_G gives the direction in which $\sigma_{\theta,\max}$ occurs. The branches will propagate in their initial directions only when $\beta = \beta_G$, but they will repel or attract each other according as $\beta < \beta_G$ or $\beta > \beta_G$.

This characteristic angle β_G depends on b/a as shown by Table 4 and Fig. 15. β_G decreases rapidly and then gradually with increasing values of b/a , and seems to approach about 10° for long branches. $(\beta_G)_{b \rightarrow 0} = 27.3^\circ$ may seem too large compared with previous experiments, but this is just a mathematical limiting value and not a practical one. In actual cases, propagation and branching of a crack will occur not by an infinitesimal length but by some finite length due to material inhomogeneities and finite grain size. If the average branching angle 2β in brittle fracture and stress corrosion cracking is $30^\circ \sim 40^\circ$ [4], it corresponds to β_G for $a/b = 0.1 \sim 0.02$ as shown in Table 4.

We found that β_G decreases as b/a increases. Therefore, if secondary branching does not occur, the branches tend to bend inward during propagation, and if secondary branching takes place, the mean of the new two directions will be smaller than β_G . The above behaviors were observed in the experiments by Kalthoff [2] and the analysis by Pärletun [8].

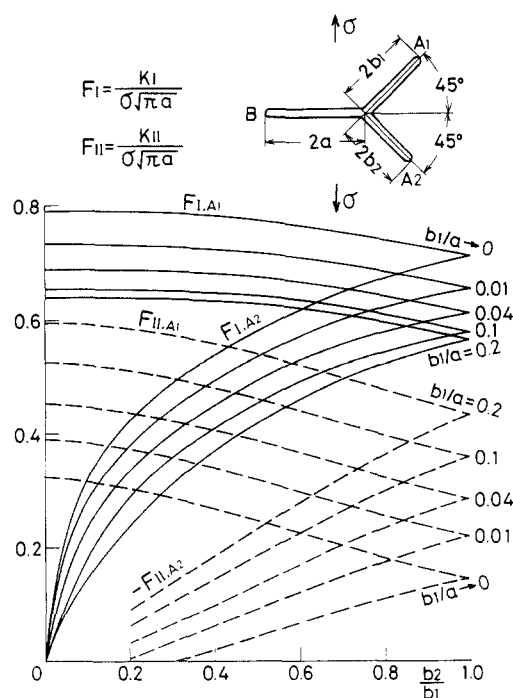


Fig. 16. Crack with unequal branches subjected to tension ($\beta_1 = -\beta_2 = 45^\circ$, $b_1 \neq b_2$).

Table 5. Limiting values when $b_1, b_2 \rightarrow 0$ in the problem shown in Fig. 17

b_2/b_1	F_{I,A_1}	F_{I,A_2}	F_{II,A_1}	$-F_{II,A_2}$
1.0	0.712	0.712	0.146	0.146
0.8	0.740	0.676	0.183	0.109
0.6	0.765	0.622	0.225	0.065
0.4	0.783	0.542	0.267	0.019
0.2	0.791	0.427	0.304	-0.023
0	0.791	0	0.323	0

3.4. Numerical results for other branched cracks (tension)

Figure 16 gives the results for cracks with two unequal branches inclined by 45° to the main crack. Assuming $b_1 \geq b_2$ without loss of generality, F_{I,A_1} , F_{I,A_2} and F_{II,A_1} , F_{II,A_2} for five values of b_1/a are shown by solid curves and dashed curves, respectively, in the whole range of b_2/b_1 . Table 5 shows their limiting values when $b_1, b_2 \rightarrow 0$ for various fixed values of b_2/b_1 .

Figure 17 gives the results for a branch emanating in the 45° -direction from a point on a straight crack, where F_{I,A_1} and F_{I,A_2} are plotted against b_2/b_1 for various values of b_1/a . Values of F_{I,A_1} for $b_2/b_1 = 0$ of course agree with the previous results for kinked cracks [10].

Figure 18 compares F_{I,A_1} and F_{II,A_1} for the tension of 45° -kinked crack BOA_1 with those for doubly and triply branched cracks. The solid curves, dashed curves and chained curves correspond to the above three cases, respectively, and we find that F_{I,A_1} and F_{II,A_1} decrease with additional branches.

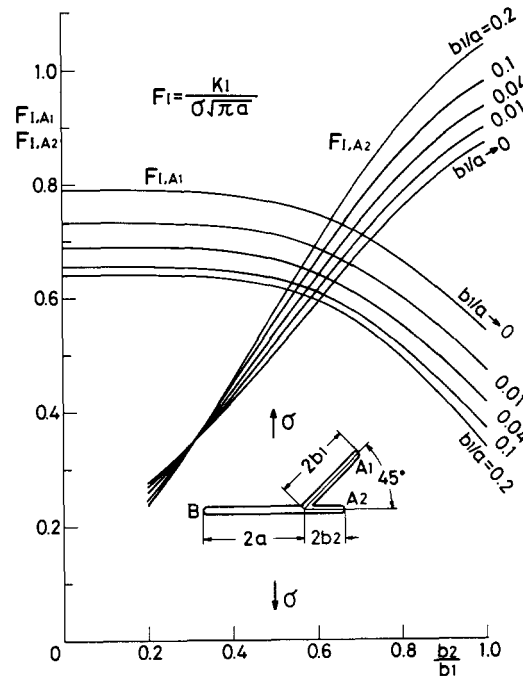


Fig. 17. Branch emanating from a point on the main crack subjected to tension.

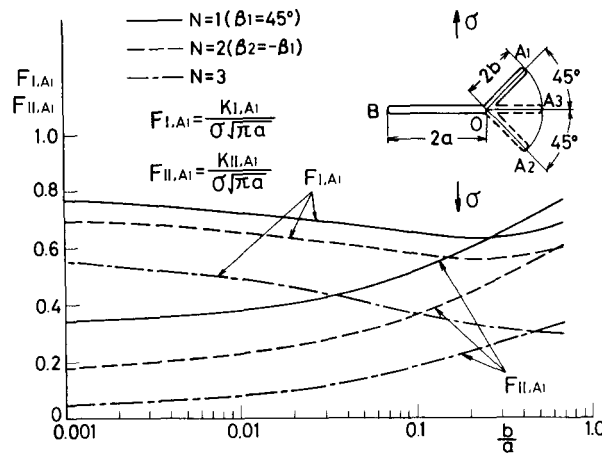


Fig. 18. Variations of $F_{I,A1}$ and $F_{II,A1}$ of 45°-kinked cracks due to additional branches.

4. Concluding remarks

Arbitrarily branched cracks in a wide plate were treated theoretically under three fundamental loadings at infinity. The analysis was performed by the body force method combined with a perturbation procedure, and the dimensionless stress intensity factors were given by power series of $\lambda = b/a$, where $2a$ and $2b$ are the lengths of the main crack and one of the branches.

Emphasis was put on the cracks with two symmetrical branches. Power series formulae with

respect to the branching angle β and dimensionless branch length λ were proposed for three fundamental loads. They are considered to give very accurate values in the range where $10^\circ \leq \beta \leq 90^\circ$ and $\lambda \leq 0.3$. Formulae for arbitrary uniform loads at infinity are given just by superposing those for the above fundamental cases.

Numerical results for various symmetrical and unsymmetrical branched cracks are given and discussed.

Acknowledgements

Authors wish to express their thanks to Professor K.B. Broberg for his invaluable advice in preparing the manuscript.

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